

Radicals, Rational Exponents, Domains, and Approximate Solutions

Rational exponents, combining like radicals, solving radical equations with domain and extraneous-root checks, and approximating irrational solutions

The rules this sheet runs on.

- $a^{m/n} = \sqrt[n]{a^m}$ — the denominator is the root, the numerator is the power.
- Radicals combine like terms only when the radicands match after simplifying: $2\sqrt{3} + 5\sqrt{3} = 7\sqrt{3}$, but $\sqrt{2} + \sqrt{3}$ does not combine.
- Squaring both sides of an equation can **create** solutions that the original equation does not have. Every candidate must be substituted back.
- $\sqrt{}$ returns the **non-negative** root, so an equation whose other side is negative has no solution there.

Directions.

- Decide first what kind of task it is: rewrite, combine, solve, or approximate. The four kinds alternate.
- On every equation, state the domain restriction and check each candidate.
- Round approximations to two decimal places unless told otherwise.

1. Evaluate without a calculator: $49^{1/2}$

Say in one sentence what the exponent $\frac{1}{2}$ instructs you to do.

Answer: _____

2. Write $\sqrt{50}$ in simplest radical form.

Find the largest perfect square that divides 50, then take its root outside the radical.

Answer: _____

3. Solve for x : $\sqrt{x+7} = 4$

(a) State the domain: for what values of x is the left side defined?

(b) Solve, then substitute your answer back into the original equation to confirm it works.

Answer: _____

4. Approximate $\sqrt{20}$ to two decimal places.

Simplify the radical first, then approximate. Show the simplified exact form as well as the decimal.

Answer: _____

5. Evaluate without a calculator: $27^{2/3}$

Take the root indicated by the denominator before applying the power in the numerator — the numbers stay small that way.

Answer: _____

6. Combine into a single radical term:

$$3\sqrt{12} + \sqrt{27}$$

Neither radical is in simplest form yet, so simplify both before deciding whether they are like terms.

Answer: _____

7. Solve for x : $\sqrt{2x+3} = x$

(a) Square both sides and solve the resulting quadratic. You should get two candidates.

(b) Test both candidates in the *original* equation. Put the surviving solution on the answer line, and explain in one sentence why the other one fails.

Answer: _____

8. Approximate $\sqrt{2} + \sqrt{8}$ to two decimal places.

Simplify and combine the two radicals into a single term first. Then say why $\sqrt{2} + \sqrt{8}$ is *not* $\sqrt{10}$.

Answer: _____

9. Evaluate without a calculator: $16^{-1/2}$

The exponent is both negative and fractional. Handle one at a time, and state whether the result is positive or negative before you compute it.

Answer: _____

10. Solve for x : $\sqrt{x+5} = x-1$

(a) Before squaring, state what must be true of $x-1$ for a solution to exist at all.

(b) Square, solve, and test both candidates in the original equation. Put the surviving solution on the answer line.

Answer: _____

11. Write as a single power of x :

$$x^{1/2} \cdot x^{1/3}$$

The bases match, so the exponents add. Give the answer as x to a single rational exponent.

Answer: _____

12. Challenge. Two equations that look almost the same:

$$(i) \quad \sqrt{2x+7} = x+2 \qquad (ii) \quad \sqrt{2x+7} = x+1$$

(a) Solve equation (i). Square, solve the quadratic, test both candidates, and put the surviving solution on the answer line.

(b) Now solve equation (ii). Its exact solutions come out as $\pm\sqrt{6}$. One is extraneous — say which and why, then give the surviving solution rounded to two decimal places.

(c) In one sentence, explain why squaring is what makes the extraneous-root check necessary in both parts.

Answer: _____